

Stale by Construction: A Logical-Floor Violation in Compound Prediction-Market

Contracts

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1. Introduction

A prediction-market contract is a binary claim that pays \$1.00 if an outcome occurs and \$0.00 if it does not; before settlement it trades between those bounds at whatever price buyers and sellers agree reflects the outcome's probability. Polymarket US launched regulated, real-money contracts of this kind in January 2026, and its coverage of the 2026 FIFA World Cup listed, for each tracked player in each match, three separate contracts on a central limit order book: at least one goal, at least one assist, and at least one goal-or-assist. Each has its own order book and its own update cadence.

These three contracts are not independent instruments. Scoring a goal is one of the ways to record a goal-or-assist, so the compound contract can never be worth less than either single-stat leg:

$$P(\text{goal} \cup \text{assist}) \geq \max(P(\text{goal}), P(\text{assist})).$$

When the compound contract is quoted less actively than the legs that decompose it, its price drifts below that floor. This is a pricing error the market makes against itself, observable without reference to any position taken. Across 157,595 simultaneously captured three-leg quotes, the compound bid falls below the goals bid in 28.81% of pre-kickoff observations, and the compound contract carries a $3.81\times$ wider spread and rests $5.6\times$ less size than the goals leg it can be priced against.

Whether that staleness reverts is a separate question, and it is tested rather than assumed. Conditioning on positions entered while the compound contract was cheap relative to its formula fair value, the mispricing narrows toward fair value 63.0% of the time, against 15.1% under a volatility-matched random walk ($z = 6.94$, $p < 0.000001$). A second strategy trades confirmed public events directly: when a goal, a substitution, or a final whistle makes one side of a contract near-certain, a resting order on the other side has not yet requoted, and whoever acts before the

book adjusts is trading a latency gap.

The purpose of this paper is to measure the pricing inefficiency, test whether it is a consistent and statistically significant effect, and evaluate whether either strategy built around it beats the market's own pricing rather than merely trading faster than it.

2. Data Pipeline and Screening Architecture

The analysis relies on a fully automated data pipeline that identifies, validates, and tracks pre-kickoff prop markets while minimizing contamination from fundamentally driven repricing. Its objective is to construct a clean sample of simultaneously observed three-leg quotes and to measure price behavior in the window where the compound contract de-stales.

The pipeline runs unattended on Windows Task Scheduler and integrates three data sources. First, market and event data is read from Polymarket US's public gateway API, which carries a live score and match clock on each event object. Second, the account's own fills, balances, and settlements are read from the authenticated portfolio API, signed with an Ed25519 key over the message `{timestamp}{method}{path}`. Third, event timestamps used for latency measurement are drawn from neutral match reports (FIFA, ESPN, Al Jazeera), never from Polymarket's own price data, so that the latency figures are not circular.

Polling is tiered by time to kickoff: every twenty minutes beyond six hours out, tightening to every sixty seconds inside the final thirty minutes. Market discovery is structural, not a hand-maintained list. The set of soccer leagues is pulled live from the platform's own sports endpoint, which returned seventy leagues besides the World Cup as of 2026-08-08; each league's games are then enumerated directly against the API.

A clean three-leg observation is recorded only when the goals, assists, and goals-or-assists contracts for the same player can all be read from the same pre-kickoff market state in a single pass. Pairing all three legs at identical capture timestamps yields 157,595 exactly-simultaneous observations, of which 124,571 fall in the forty-eight hours before kickoff. This is the sample used throughout the paper.

Every fill the account made is reconciled at ingestion: each trade row's quantity is checked

against its cost divided by its price, and the local trade log is cross-checked one-to-one by trade identifier against a fully paginated pull of the account's complete live history (392 trades). The trading client used for these reads exposes only balance, position, and activity lookups; it contains no order-placement methods.

3. Signal Identification

For a given player in a given match, let g , a , and ga be the traded prices of the “1+ goals,” “1+ assists,” and “1+ goals-or-assists” contracts. Under a statistical-independence approximation, inclusion–exclusion gives a fair value for the compound contract:

$$P(\text{goal} \cup \text{assist}) = P(\text{goal}) + P(\text{assist}) - P(\text{goal} \cap \text{assist}) \approx g + a - ga.$$

The signal is the difference between that fair value and the contract's actual price:

$$gap = (g + a - ga) - ga.$$

A positive gap means the compound contract is trading cheap relative to its own legs, which is a buy signal. Across 208 clean pre-kickoff samples, gap averaged +0.0106 and exceeded 0.05 in absolute value in 31 of 208 cases — a meaningful mispricing on an instrument priced between \$0.00 and \$1.00.

The reference constraint is the logical floor stated in Section 1: because a goal is one route to a goal-or-assist, a correctly priced compound contract can never trade below either single-stat leg. A violation of that inequality is a pricing error that requires no model to identify — only three simultaneous quotes.

This is not riskless arbitrage. The independence approximation carries model risk in both directions: a goal is negatively correlated with the same player's own assist, which pushes true $P(\text{goal} \cap \text{assist})$ below ga , while a player heavily involved in attack has elevated odds of both outcomes for reasons the formula cannot see. The strategy is therefore statistical relative-value

arbitrage with adverse-selection exposure: a resting bid improved toward fair value can still be filled by a counterparty more informed than the formula.

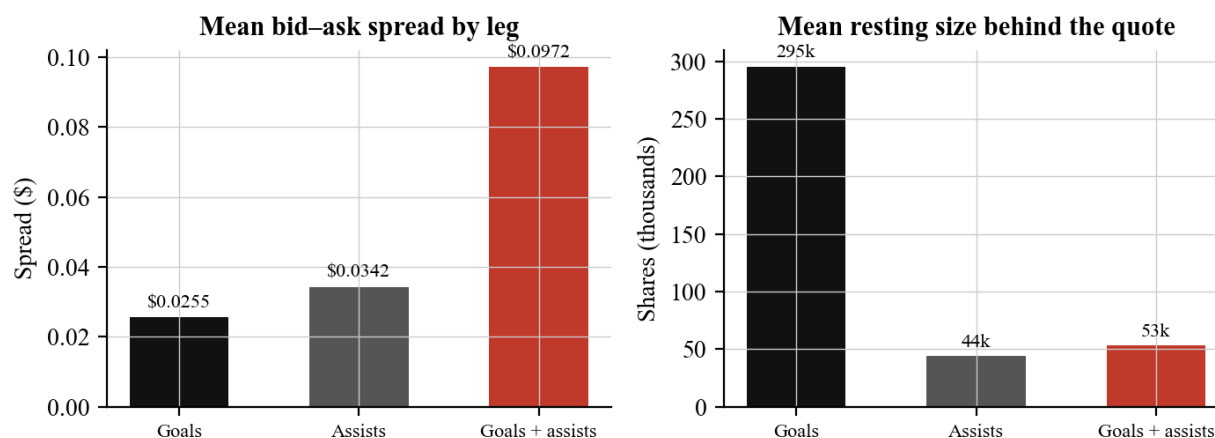
4. Data Analysis

Every snapshot of quotes and book depth is bucketed by time to kickoff. The four figures below test, directly against the data, the mechanisms the two strategies depend on.

Spread and Resting Depth by Market Leg

The plot below displays the mean bid–ask spread and the mean resting size for each of the three legs, measured over the forty-eight hours before kickoff for the fourteen highest-coverage players.

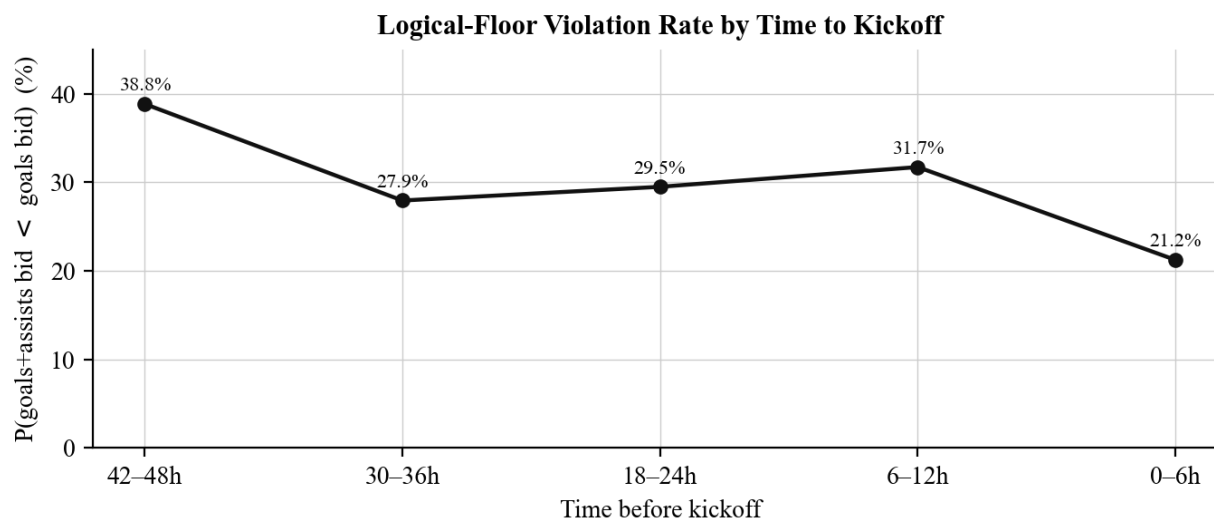
Spread and Resting Depth by Market Leg, 48h Pre-Kickoff



The plot shows that the compound contract is structurally the stale one, on both measures at once. It carries a $3.81\times$ wider spread than the goals leg (\$0.0972 against \$0.0255) and rests $5.6\times$ less size behind its quotes (52,966 against 295,156 shares), widening to $7.5\times$ in the final four hours. The goals leg is the anchor: its spread is essentially flat near \$0.019 for the entire two days, and its book grows from roughly 295,000 to 376,000 shares as retail attention arrives. The compound market behaves nothing like it — it opens at a \$0.157 spread and compresses to \$0.047 by kickoff, but its resting size never grows at all. Retail flow arrives into the simple market and largely ignores the compound one.

Logical-Floor Violation Rate by Time to Kickoff

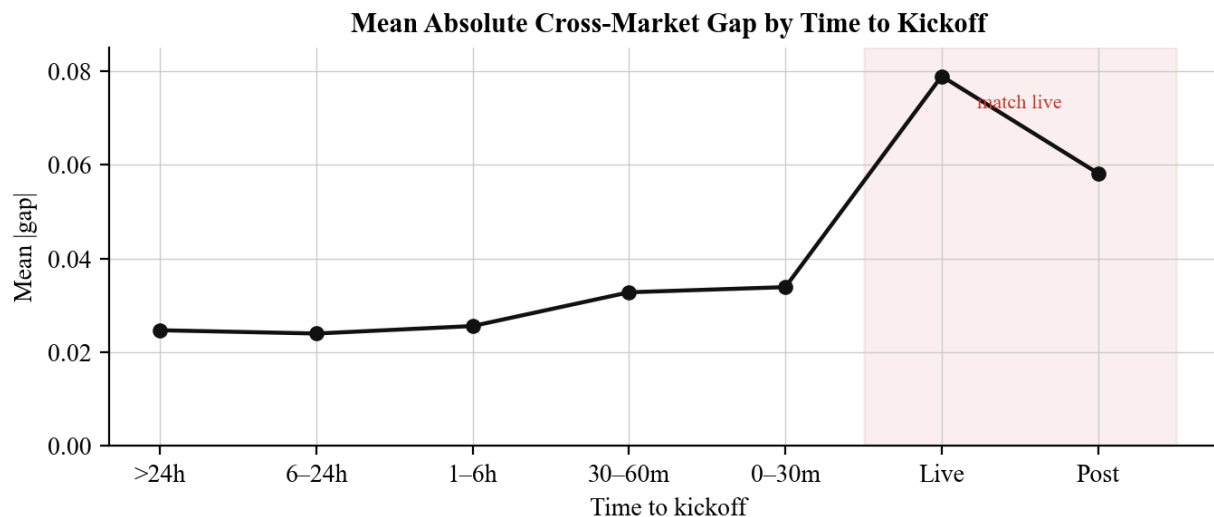
The plot below displays the fraction of pre-kickoff observations in which the compound bid sat below the goals bid, grouped by hours before kickoff.



The plot shows that the best resting bid on the strictly-more-likely outcome sat below the best resting bid on a subset of it in 28.81% of pre-kickoff observations (35,894 of 124,571), by a mean of \$0.088. The violation is highest far from kickoff — 38.83% at forty-two to forty-eight hours out — and decays to 21.23% inside the final six hours, exactly as a de-staling process predicts. It also runs opposite to fame: Lionel Messi’s compound book violated the identity six times in 7,287 observations, while players one tier below the household names — famous enough for their markets to attract flow, not so famous that the compound leg was watched as closely as the simple one — carried violation rates above 20%.

Mean Absolute Cross-Market Gap by Time to Kickoff

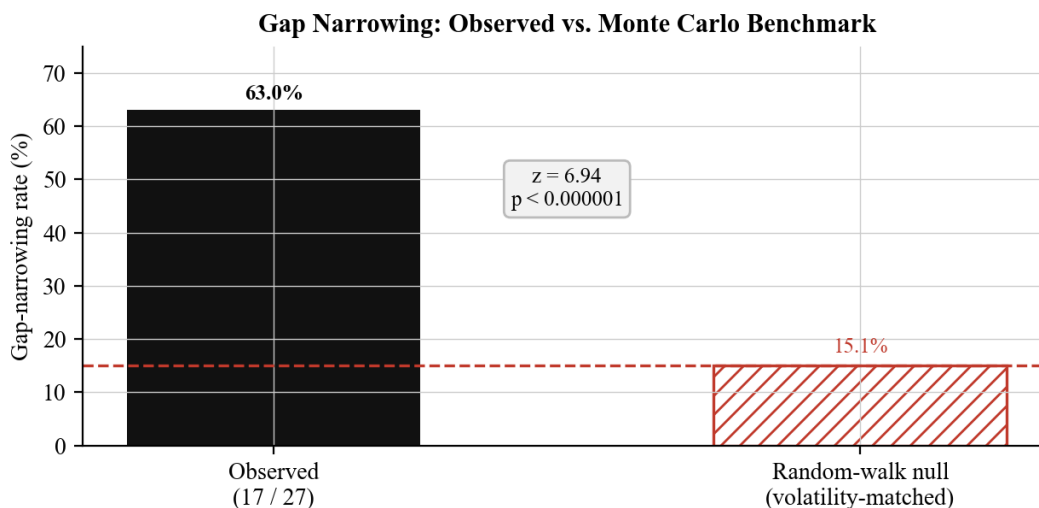
The plot below displays the mean absolute difference between the compound price and its formula fair value, by time bucket, from more than a day before kickoff through live play.



The plot shows that the gap is roughly flat near 0.025 for the entire pre-kickoff window and then more than triples, to 0.079, once the match goes live. Real scoring events hit the three legs at different speeds, and the resulting lag is the inefficiency the second strategy trades.

Gap Narrowing: Observed vs. Monte Carlo Benchmark

To test whether the pre-kickoff mispricing actually reverts rather than merely drifting, per-step increments of gap were computed from 182,760 consecutive same-player, same-game observations (median step about sixty-six seconds), giving a mean of approximately zero and a standard deviation of $\sigma = 0.0227$ per step. For each of the twenty-seven cleanly pre-kickoff-exited positions with a resolvable gap reading at both entry and exit, 5,000 random-walk paths were simulated — each starting at that position's actual entry gap and taking its actual number of observed steps — and the fraction of paths ending with a smaller absolute gap than they started with was recorded.



The plot shows that the observed data narrows far more often than the volatility-matched benchmark: 63.0% of the real positions closed toward fair value, against 15.1% of the simulated paths. Because the simulation is calibrated directly from the real second-by-second movements and assumes no drift, the difference reflects structure in the price process rather than a parameter choice.

5. Statistical Testing

Each test below is a one-sample proportion z -test. Where the null probability is an entry price rather than a fixed benchmark, the test asks whether the strategy's realized results differ from what the market's own pricing implied — the correct question for a strategy that may simply be fast rather than better informed.

Test 1: Is the gap mean-reversion real?

The null hypothesis is that the gap signal follows a driftless, volatility-matched random walk, which narrows 15.1% of the time.

$$H_0 : p = 0.151 \quad H_1 : p > 0.151 \quad n = 27, \quad \hat{p} = 0.630$$

$$z = \frac{0.630 - 0.151}{\sqrt{\frac{0.151(1 - 0.151)}{27}}} = 6.94, \quad p < 0.000001.$$

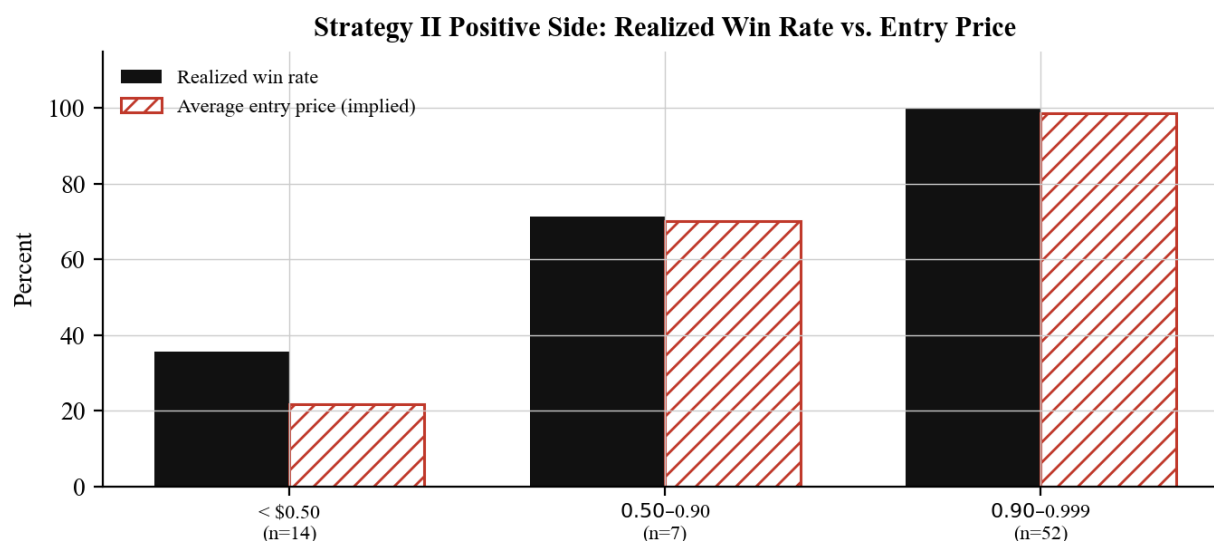
Because the p -value is effectively zero, we reject the null hypothesis. The one caveat worth stating: contract prices live in $[0, 1]$, so gap is bounded while the simulated walk is not, and a bounded process starting far from zero must revert eventually simply because it cannot escape. Two things bound how much that can explain. The effect size is large — 63.0% against 15.1% is not a margin a boundary artifact plausibly manufactures at these gap magnitudes — and the low simulated rate is driven by the walk having room to drift over a typical holding period, not by the walk being pushed outward. Re-running the null with reflecting barriers at the price bounds would raise the simulated rate somewhat but would not close the gap.

Test 2: Does the speed strategy's positive side beat the market's pricing?

The positive side buys “Yes” on a confirmed goal or assist. The null hypothesis is that the market was already pricing those contracts efficiently, so the true win probability equals the average entry price.

$$H_0 : p = 0.812 \quad H_1 : p \neq 0.812 \quad n = 73, \quad \hat{p} = 0.849$$

$$z = \frac{0.849 - 0.812}{\sqrt{\frac{0.812(1 - 0.812)}{73}}} = 0.81, \quad p = 0.21.$$



We fail to reject the null hypothesis. The plot shows the realized win rate tracking the

average entry price closely in every bucket — the market was already priced fairly. The dominant bucket by volume — 52 of 73 positions — resolved “Yes” 100% of the time against a 98.7% average entry price, which is exactly what a well-calibrated market predicts. The positive side’s profit is real and mechanically reliable, but it is not evidence of beating the market’s pricing: the edge is being fast enough to buy before the book reprices, not being smarter about the true probability once bought.

Test 3: Is the speed strategy’s negative side calibrated?

The negative side buys “No” on a confirmed substitution or full-time whistle — mechanically a sell of “Yes.” The null hypothesis is that the market was already pricing those contracts efficiently, so the true win probability equals the average implied “No” price.

$$H_0 : p = 0.956 \quad H_1 : p \neq 0.956 \quad n = 102, \quad \hat{p} = 0.892$$

$$z = \frac{0.892 - 0.956}{\sqrt{\frac{0.956(1 - 0.956)}{102}}} = -3.15, \quad p = 0.0016.$$

We reject the null hypothesis, in the unfavorable direction: the negative side’s realized results are worse than its own entry prices implied. The shortfall lives entirely in the thirteen positions entered below 90% implied confidence, where a 30.8% realized win rate ran against a 76.4% implied one. Two positions drive it, and Section 6 examines them directly. The dominant, highest-confidence majority (89 of 102 positions) is calibrated almost exactly as well as the positive side.

Multiple comparisons and in-sample validity

Bonferroni-correcting the three tests above (threshold $0.05/3 = 0.0167$) changes no conclusion: Test 1 and Test 3 clear the corrected threshold comfortably, and Test 2 is non-significant either way. The larger limitation is that the gap signal was designed and evaluated on the same thirty-five-day dataset, with no out-of-sample holdout. A complete census of every soccer competition Polymarket US lists finds that none of the other seventy leagues currently carries the three-parallel-market structure the signal requires, so a genuine replication needs a second tournament that does

not yet exist. The collector already discovers events across all seventy leagues, so that data begins accumulating the moment one lists a player goal or assist market again.

6. Risk

A key risk to the patience strategy arises when the compound contract's staleness does not close before kickoff. The exit is a passive resting sell, not a market order — it depends on a counterparty's order arriving — and when it does not fill in time the position is carried into live play, where match variance layers on top of an already-late exit. Of 192 round-trip positions, only 56 fully exited before kickoff as intended; the remaining 136 were carried into the match. The 56 clean exits won 94.6% of the time; the 136 delayed positions won only 78.7%, with lower average returns — a real, quantified cost of fill risk, though the clean-exit figure is partly a selection artifact, since a position moving the wrong way is more likely to still be open at kickoff than sold at a loss beforehand.

Jeremy Doku's goals-or-assists position ahead of Belgium–Senegal is a clear example. It was bought across three pre-kickoff fills at \$0.29 to \$0.31, a blended \$0.2978, and was not sold before kickoff. Carried into the match, it filled across six tranches at \$0.38 to \$0.41, realizing +\$25.18 (+27.6%) and +\$23.57 on its two largest tranches. The mispricing it was entered on was real; it simply closed late, and a first move the other way would have produced the opposite result on the same delay.

The speed strategy's negative side carries a different risk: a confirmed substitution or full-time whistle caps a player's remaining probability, but two positions were entered well short of certainty and resolved against the call. Charles De Ketelaere (goals-or-assists, Spain–Belgium, 07-10), entered at an average “No” price near 79%, lost \$73.47 — the largest single-position loss in the strategy data. Jovo Lukić (goals, Bosnia and Herzegovina–Qatar, 06-24), entered near 88%, lost \$46.38. Both are genuine signal misses rather than execution artifacts, and together they account for the negative side's calibration shortfall in Test 3. The remaining fifty-two below-90% entries and the entire above-90% population resolved as their prices implied.

To keep the latency measurement honest, event timestamps are taken only from neutral

match reports, never from Polymarket's price data, and second-half goals are excluded rather than approximated: converting a second-half minute to a wall-clock time requires the real halftime-break length, and assuming a standard fifteen minutes can swing a computed latency by 300 seconds — more than the entire quantity being measured.

7. Execution Model

Pre-Trade Identification and Filtering

- **Clean three-leg observation.** A candidate surfaces only when all three legs quote from the same pre-kickoff market state and the gap exceeds a threshold; $|gap| > 0.05$ held on 31 of 208 clean samples.
- **Star-player universe.** Positions concentrate on high-coverage names — Messi, Mbappé, Haaland, Kane and similar — whose compound markets de-stale reliably enough to be exited into before kickoff. An obscure player can carry the same theoretical gap with far less confidence it closes and far less volume to exit into.

Entry Logic

- **The fair-value ceiling.** Rather than crossing the spread, the patience strategy repeatedly improves its own resting bid in small increments, capped strictly below the formula-implied fair value, so the position buys at or below its own fair-value estimate even in the best case for a counterparty. On the fills, 96.2% of pre-kickoff compound buys were priced below the formula value computed from the concurrent leg asks.
- **Passive only.** Section 4 shows the inefficiency sat almost entirely on the bid side; a crossable, locked arbitrage — buying the compound at its ask and selling the goals leg at its bid — existed in only 0.05% of observations. An aggressive strategy would have found nothing to take.
- **The confirmed-event trigger.** The speed strategy does not use price level as a trigger. On a confirmed goal, assist, substitution, or final whistle for a tracked player, the near-certain side is bought immediately at whatever price is showing and held to \$1.00 settlement.

Risk and Position Management

- **Fixed-dollar sizing.** Positions are sized as a fixed fraction of available capital rather than a share count, for consistency across contract prices.
- **Hard cap.** Total portfolio risk is capped at 5% of capital, checked at kickoff; a position built past that fraction across multiple fills is cut back with a below-market sell.
- **Breadth over size.** The speed strategy spreads capital across every still-unresolved candidate at the moment of the trigger rather than concentrating on one, since the confirming event makes each outcome close to certain rather than merely likely.

Exit Strategy

- **Staged sell-off.** Inside the final five minutes before kickoff, the patience strategy sells roughly a quarter of whatever remains once per minute, crossing the spread each round to guarantee that tranche fills.
- **Unconditional backstop.** A final sweep at kickoff itself closes anything the schedule leaves behind, so the position is always flat by kickoff — the property the trade record found missing in 136 of 192 cases.
- **Hold to settlement.** Once a speed-strategy entry lands near \$0.98 to \$0.99, the remaining margin is too thin, net of the trading fee, to be worth an active sell. Zero of 192 patience-strategy round-trips were ever sold at a price at or above \$0.90; every position that reached that band was held to settlement.

The live order-placement layer for the patience strategy was developed jointly with a collaborator on a separate account and is described here as reconstructed from that logic and cross-checked against the surviving trade ledger; every figure in this paper derives from the author's own account and its own records.

8. Final Weights

Two execution schedules follow from the model above. The pre-kickoff exit, which exists to close the fill-risk gap in Section 6, sells a fixed fraction of the remaining position each minute

inside the final five, with an unconditional backstop at kickoff:

Minutes to kickoff	Action	Fraction of remainder	Cumulative % closed
5	Staged sell	25%	25.0%
4	Staged sell	25%	43.8%
3	Staged sell	25%	57.8%
2	Staged sell	25%	68.4%
1	Staged sell	25%	76.3%
0	Unconditional sweep	100%	100%

Entries are not scheduled the same way — they are placed as the signal presents itself across the full pre-kickoff window. Of 86 pre-kickoff compound buys, the median fill landed 8.7 hours before kickoff (mean 14.6 hours), and the distribution has two human-scale clusters rather than a flat profile:

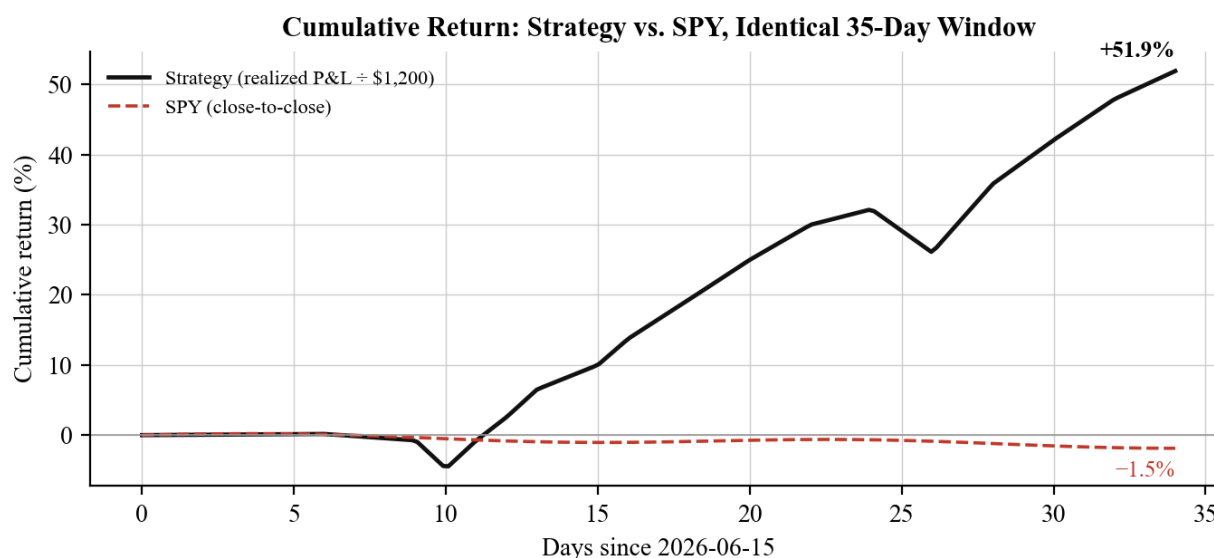
Window before kickoff	Share of pre-kickoff compound buys
18–24 hours	20%
6–18 hours	37%
Final 6 hours	43%

9. Results

The strategies were run against Polymarket US's World Cup player-prop markets over a thirty-five-day window, 2026-06-15 to 2026-07-19. Net capital contributed was \$1,200 across four completed deposits; the account carries zero open positions at close, so every figure reported is fully realized. Trading-attributable profit totaled \$691.58 — \$303.31 from the patience strategy (actively sold), \$319.72 from the speed strategy (both sides, held to settlement), and \$68.55 from non-core bet types and one settlement outside the window.

Against SPY over the identical calendar window — real closes \$754.83 on 2026-06-15 to \$743.29 on the last trading day before the window ended — the account returned +51.9%

on contributed capital from the prop strategies alone, against SPY's -1.5% . The comparison establishes only that the return is not an artifact of a broad-market tailwind.



The three strategy legs behave differently, and only one carries a statistically validated edge:

	Patience (sold)	Speed, positive (held)	Speed, negative (held)
Positions	192	62	102
Win rate	83.3%	85.5%	89.2%
Net P&L	+\$303.31	+\$126.19	+\$193.52
Validated edge?	Yes ($p < 0.000001$)	No ($p = 0.21$)	No — underperforms ($p = 0.0016$)

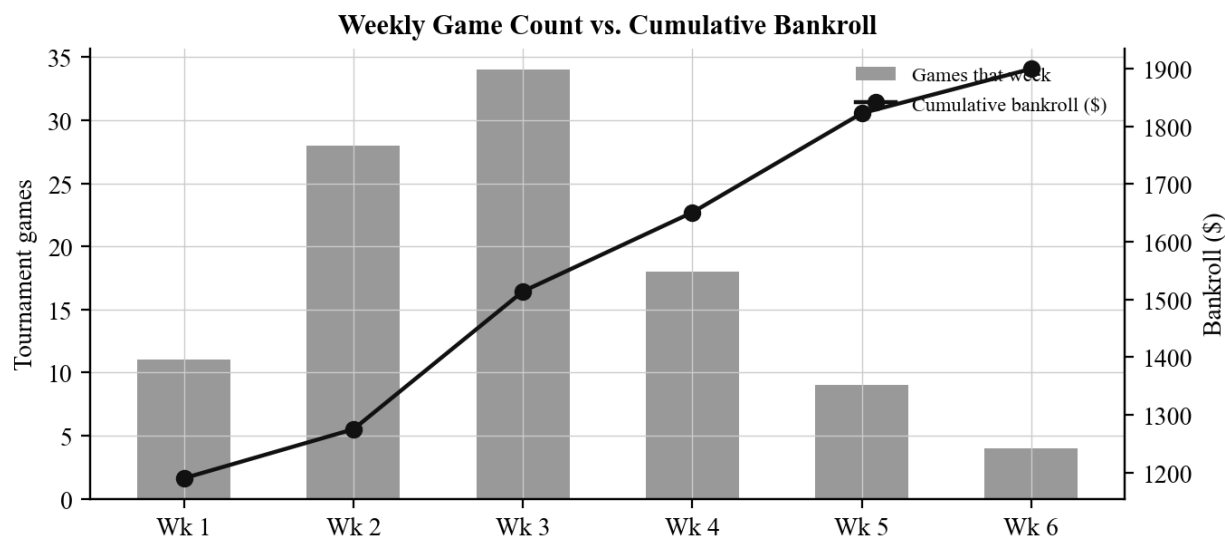
Within the patience strategy, the fill-risk split is the more informative comparison: the 56 positions that exited before kickoff won 94.6% of the time, and the 136 carried into live play won 78.7%.

Metric	Strategy	Benchmark (SPY)
Cumulative return, 35 days	+51.9%	-1.5%
Sharpe ratio, daily ($n = 31$)	0.42	—
Maximum drawdown	\$73.94 (11.9% of peak)	—

The annualized Sharpe is deliberately not reported. Thirty-one daily observations from a single bounded event put a wide confidence interval on the daily estimate, and scaling that interval by $\sqrt{365}$ produces a range broad enough to be uninformative while presenting as precision. Restricting the calculation to the period after all \$1,200 was deposited gives a daily Sharpe of 0.57 on $n = 23$ — higher, not lower, once the low-capital ramp-up is removed.

Capacity

The clearest single fact in the data about whether the approach generalizes is that capital grew exactly as opportunity shrank.



Tournament game count fell from 11 to 28 to 34 to 18 to 9 to 4 per week as the bracket narrowed, while the account's bankroll rose monotonically from roughly \$1,190 to \$1,823 across those same weeks. The account held the least capital when opportunity was greatest and the most when it was smallest. Combined with measured top-of-book depth of 1,406 to 9,762 shares against an average position size of \$17.98, and a median thirty-eight-minute lag between a game-ending event and settlement freeing the capital, the strategy is capacity-bound by construction.

10. Conclusion

This study demonstrates that a thinly-quoted compound prediction-market contract, priced against the more actively quoted single-stat contracts that decompose it, carries a quantifiable and

mechanical pricing inefficiency. Its bid falls below the logical floor $P(\text{goal} \cup \text{assist}) \geq P(\text{goal})$ in 28.81% of 124,571 pre-kickoff observations, and that staleness reverts toward fair value far more often than a volatility-matched random walk predicts — 63.0% against 15.1%, $p < 0.000001$. A second strategy, trading confirmed public events directly, is profitable but mechanical: a calibration test finds its positive side statistically indistinguishable from its own entry pricing, and its negative side significantly worse. The account converted \$1,200 into \$691.58 in fully realized profit over thirty-five days, outperforming SPY (+51.9% against -1.5%) over the identical window.

The inefficiency is real, but it is capacity-bound by construction. The account held the least capital when tournament opportunity was greatest and the most when it was smallest, and measured book depth against realized position size leaves no room to scale without redesigning the execution layer. Ultimately, the findings suggest that a retail-heavy prediction market during a marquee event prices its simple contracts efficiently and its compound contracts against itself — an asymmetry a faster, better-modeled counterparty can capture at small size, but not, on this evidence, at institutional scale.